

IrisGuard: An AI-Powered Iris Biometric Authentication Framework with Multi-Phase Presentation Attack Detection, Explainable AI, and Open-Set Recognition

Research Project Summary

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Abstract—Iris recognition offers strong biometric accuracy, but practical deployment is hindered by two recurring weaknesses: susceptibility to presentation attacks (e.g., printed photos or screen replays) and a lack of transparency in how AI-based decisions are reached. This brief presents *Iris Guard*, a cross-platform mobile biometric authentication system organized as a multi-stage authentication pipeline covering iris detection, liveness/presentation-attack detection with visual explainability, and identity verification. The system is delivered as a Flutter application with a companion administrative console and an integrated conversational assistant. In internal evaluation, the identity verification stage reached an AUC of 0.997 and an Equal Error Rate of 3.25%, exceeding the project’s own target thresholds. Full methodology, dataset composition, and model training details are withheld pending completion of the associated research and are not included in this summary.

Index Terms—biometric authentication, iris recognition, presentation attack detection, explainable AI, mobile computing

I. INTRODUCTION

Biometric authentication is increasingly relied upon in mobile and security-critical systems because it binds identity verification to a physical trait rather than a memorized secret. Iris texture, in particular, offers high inter-subject distinctiveness and long-term stability. However, three practical gaps limit real-world adoption: (1) naive iris capture pipelines are vulnerable to presentation attacks such as printed or replayed images; (2) deep-learning-based security decisions are typically opaque, offering no human-interpretable justification; and (3) systems must reliably reject unenrolled, unknown identities (open-set recognition) rather than only matching against known users. Iris Guard is designed to address these three gaps within a single, mobile-deployable pipeline.

II. SYSTEM DESIGN

Iris Guard structures every authentication attempt as a sequence of three independently gated phases, so that a failure or rejection at any stage halts the attempt before it can proceed further (Fig. 1). This defense-in-depth structure is intended to keep failure modes localized and auditable.

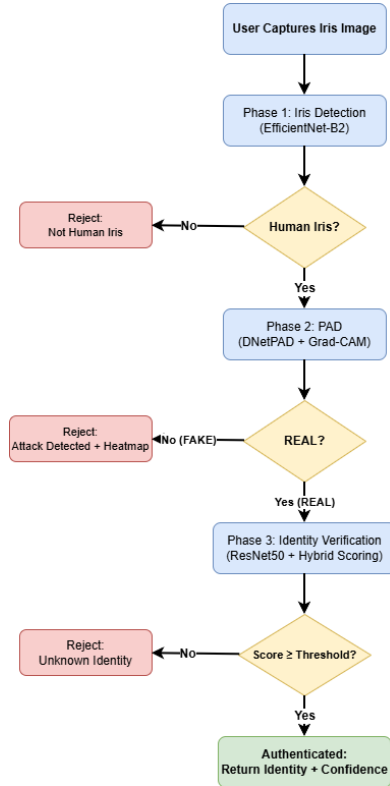


Fig. 1: High-level authentication workflow used by Iris Guard.

A. Phase 1: Iris Detection

The captured image is first screened to confirm that it genuinely contains a human iris before any downstream processing is triggered, preventing wasted computation and reducing exposure of later stages to malformed or irrelevant input.

B. Phase 2: Liveness and Presentation Attack Detection

The system determines whether the presented iris corresponds to a live capture or a spoofing attempt (e.g., printed

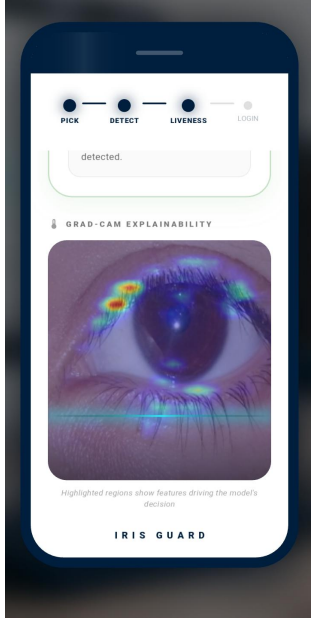


Fig. 2: Example visual explanation generated for a liveness decision.

photo, screen replay). Uniquely, this phase produces a visual saliency overlay (Fig. 2) highlighting the image regions that most influenced the liveness decision, giving users and auditors an interpretable justification rather than an opaque confidence value alone.

C. Phase 3: Identity Verification

Only iris samples that pass both the detection and liveness gates proceed to identity verification, where the sample is matched against enrolled identities and a confidence-scored accept/reject decision is returned.

D. Application Layer

The pipeline is exposed through a cross-platform Flutter application covering: multi-image user enrollment, real-time authentication with step-by-step status feedback, an administrative console for monitoring active users, scan volume, and system health, and an integrated conversational assistant to guide users through enrollment, verification, and troubleshooting.

III. EVALUATION SUMMARY

Table I summarizes headline performance of the identity verification stage against the project’s internal target thresholds. Fig. 3 shows the corresponding ROC curve.

All six tracked metrics exceeded their internally defined targets, indicating strong separation between genuine and impostor authentication attempts under the tested conditions.

IV. DISCUSSION AND SCOPE

This brief intentionally reports system-level architecture and headline outcomes only. Model architectures, training procedures, dataset composition, and detailed experimental protocol are omitted, as the associated research is still in

TABLE I: Headline Performance Metrics

Metric	Achieved	Target	Status
True Accept Rate (TAR)	91.48%	$\geq 88\%$	Pass
True Reject Rate (TRR)	96.79%	$\geq 95\%$	Pass
False Accept Rate (FAR)	3.31%	$\leq 5\%$	Pass
False Reject Rate (FRR)	8.36%	$\leq 10\%$	Pass
Equal Error Rate (EER)	3.25%	$\leq 3.5\%$	Pass
Area Under Curve (AUC)	0.9974	≥ 0.99	Pass

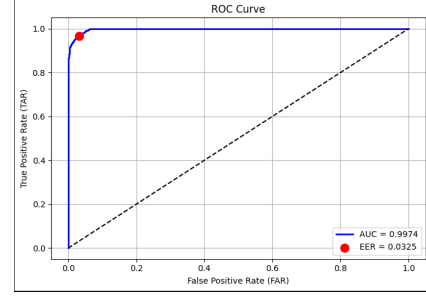


Fig. 3: ROC curve for the identity verification stage.

progress and is being prepared for formal publication. The results reported here should therefore be read as an internal evaluation snapshot rather than a peer-reviewed benchmark.

V. CONCLUSION

Iris Guard demonstrates that a mobile-deployable iris authentication system can combine layered presentation attack detection with explainable liveness decisions, without sacrificing verification accuracy. Future work will focus on expanding the open-set recognition evaluation, broadening the presentation-attack dataset, and completing the formal write-up of the underlying methodology for publication.

Research Notice. This document provides a high-level overview of the Iris Guard research project. Detailed methodologies, datasets, source code, model architectures, training procedures, and implementation details are intentionally withheld while the associated manuscript is under peer-review preparation. Unauthorized reproduction or redistribution of unpublished research materials is discouraged.

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